

A Fourth-Root Generalisation of Goursat's Algorithm

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Abstract

We extend the cyclic-Möbius framework of our previous paper [1] from cube roots to fourth roots, treating integrals of the form $\int F(t) dt/R(t)^p$ for $p \in \{1/4, 3/4\}$ where R is a quartic polynomial whose four roots admit an order-4 Möbius cyclic permutation. This last condition is equivalent to the four roots being in harmonic position, i.e., having cross-ratio -1 . Under such a symmetry, the integrand decomposes into four eigenspaces under the induced action of $\mathbb{Z}/4$, each of which descends to a curve $Y_k^{(p)}: y^4 = x^{3-k}(x-K)^{4p}$. Two of these four curves have genus 0 and yield elementary antiderivatives via explicit rational parametrizations; the remaining two are elliptic curves, all isomorphic over $\mathbb{C}$ to the lemniscate $E: y^2 = u^3 + Ku$ with j -invariant 1728, and their associated Abelian integrals are generically transcendental. The obstructive eigenspaces depend on p : at $p = 1/4$ they are V_1 and V_2 , at $p = 3/4$ they are V_0 and V_1 , exhibiting a duality $p \leftrightarrow 1-p$ in which the trivial and quasi-trivial eigenspaces V_0, V_2 swap roles. We prove the dichotomy, verify it explicitly on the canonical example $R(t) = t^4 - 1$ where each verdict is computed by hand, give a brief Mathematica implementation extending `IntegrateGoursat` to the new exponents, and discuss the lemniscate rigidity that distinguishes the case $n = 4$ among all cyclic-radical generalisations.

1 Introduction

The classical pseudo-elliptic theorem of Goursat [2], extended in our previous paper [1] to the cube-root setting, identifies elementary integrals among the family

$$\int \frac{F(t) dt}{R(t)^p},$$

in cases where R admits a Möbius automorphism cyclically permuting its roots. For square roots ($p = 1/2$, $\deg R \in \{3, 4\}$), the symmetry group is the Klein four-group V_4 acting through three involutions; for cube roots ($p \in \{1/3, 2/3\}$, $\deg R = 3$), it is the cyclic group $\mathbb{Z}/3$ acting through an order-3 Möbius transformation. The natural next case is $n = 4$.

Here a new feature emerges. For three points on the projective line, an order-3 cyclic Möbius transformation cycling them is essentially unique (a free consequence of $\mathrm{PGL}_2(\mathbb{C})$ acting triply transitively on $\mathbb{P}^1$); every cubic with simple roots therefore admits the required symmetry. For four points, however, an order-4 cyclic Möbius transformation cycling them in a prescribed order exists if and only if their cross-ratio (with that ordering) is -1 , i.e., they are arranged *harmonically* on $\mathbb{P}^1$. The fourth-root analog is therefore restricted to a codimension-1 family of quartic radicands; equivalently, we are asking the unordered 4-tuple of roots to satisfy a single algebraic equation in its j -invariant of 4 points on $\mathbb{P}^1$.

Main results

Once this hypothesis is in force, the eigenspace decomposition under the order-4 Möbius generalises cleanly. The function $H(z)$ associated to the integrand in canonical coordinates decomposes uniquely into four pieces $H = H_0 + H_1 + H_2 + H_3$ in eigenspaces V_k of the $\mathbb{Z}/4$ -action $z \mapsto iz$, and the question of elementarity reduces to vanishing conditions on certain H_k .

The reduction is genuinely richer than for $n = 3$. There, only one eigenspace (V_1) obstructed elementarity, with the other two (V_0 and V_2) leading to genus-0 reductions. For $n = 4$, two eigenspaces simultaneously obstruct, and the obstructions take the form of Abelian integrals on the elliptic curve $y^2 = u^3 + Ku$, the *lemniscate* elliptic curve with j -invariant 1728 and complex multiplication by $\mathbb{Z}[i]$. This curve is exactly the one appearing in the genus-1 rigidity remarks at the end of [1]: there it was an end-of-paper observation; here it becomes the structural backbone of the theorem.

The two exponents $p = 1/4$ and $p = 3/4$ exhibit a clean duality. At $p = 1/4$ the obstructive eigenspaces are V_1 and V_2 , while at $p = 3/4$ they shift to V_0 and V_1 . The eigenspace V_3 is always elementary; V_1 is always obstructive; V_0 and V_2 swap roles between the two exponents. Geometrically this duality is the analog of the duality between the cube-root exponents $1/3$ and $2/3$ from [1], but with one extra nontrivial eigenspace involved on each side.

Outline

Section 2 establishes the harmonic-roots condition and the canonical-form lemma. Section 3 states and proves the main theorem at $p = 1/4$. Section 4 proves the dual theorem at $p = 3/4$. Section 5 works through the canonical examples on $R(t) = t^4 - 1$, computing closed forms by hand and exhibiting both elementary and non-elementary cases. Section 6 collects concluding remarks: the rigidity of the lemniscate among $n = 4$ obstructions, the link to the bilinear duality $yy' = cx(x - K)$ between dual obstruction curves, and the algorithmic perspective.

We restrict throughout to R quartic. The case of R cubic with $n = 4$ is genuinely different — there is no order-4 Möbius cycle on three roots and infinity unless the three finite roots form an arithmetic progression — and we leave it for future work.

2 Harmonic configurations and the canonical form

The harmonic condition

Recall that the cross-ratio of four ordered distinct points $p_1, p_2, p_3, p_4 \in \mathbb{P}^1$ is

$$(p_1, p_3; p_2, p_4) = \frac{(p_1 - p_2)(p_3 - p_4)}{(p_1 - p_4)(p_3 - p_2)}.$$

The four points are said to be in *harmonic position* (with this ordering) if the cross-ratio equals -1 .

Lemma 2.1 (Existence of an order-4 cyclic Möbius). *Let $\{r_1, r_2, r_3, r_4\}$ be four distinct points on $\mathbb{P}^1$. There exists an order-4 Möbius transformation $S \in \mathrm{PGL}_2(\mathbb{C})$ with $S(r_1) = r_2$, $S(r_2) = r_3$, $S(r_3) = r_4$, $S(r_4) = r_1$ if and only if $(r_1, r_3; r_2, r_4) = -1$. When S exists, it is unique, has trace 0 in $\mathrm{PGL}_2(\mathbb{C})$, and its two fixed points α, β are distinct from the four r_i .*

Proof. A Möbius transformation S is determined by its action on three points; given the desired cyclic action on four points, S is overdetermined and exists if and only if the consistency condition is satisfied. Writing S as the unique Möbius transformation sending $(r_1, r_2, r_3) \rightarrow (r_2, r_3, r_4)$, the condition $S(r_4) = r_1$ becomes

$$(r_1, r_2; r_3, r_4) = (r_2, r_3; r_4, r_1),$$

which using the cross-ratio identity rearranges to $(r_1, r_3; r_2, r_4) = -1$.

When this holds, S has order 4 in $\mathrm{PGL}_2(\mathbb{C})$. An element of $\mathrm{PGL}_2(\mathbb{C})$ has order 4 if and only if it is conjugate to $z \mapsto iz$, i.e., has eigenvalue ratio i on a representative matrix; equivalently, its trace satisfies $\mathrm{tr}^2 = 0 \cdot \det$, i.e., $\mathrm{tr} = 0$ in PGL_2 . The two fixed points are distinct (since S is non-parabolic) and cannot equal any r_i (since S has no fixed point in the cycle). $\square$

Remark 2.2. The j -invariant of an unordered 4-tuple is the function $j(\lambda) = \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}$ of the cross-ratio λ . It is invariant under the S_3 -action on the cross-ratio that comes from reorderings preserving the unordered tuple. The harmonic value $\lambda = -1$ corresponds to $j = \frac{27}{4}$, a non-generic value distinguished as the locus where extra automorphisms (beyond the trivial S_3) appear. This is exactly the condition that allows the order-4 cyclic action.

Canonical form

Lemma 2.3 (Canonical form for quartic harmonic R). *Let $R(t) \in \mathbb{C}[t]$ be a quartic polynomial whose four roots $\{r_1, r_2, r_3, r_4\}$ are distinct and in harmonic position; let S be the order-4 Möbius transformation cycling them and let α, β be its fixed points. Define $z = (t - \alpha)/(t - \beta)$, so that S becomes $z \mapsto iz$. Then there are constants $c \in \mathbb{C}^\times$ and $K \in \mathbb{C}^\times$ such that*

$$R(t)(1 - z)^4 = c(z^4 - K). \quad (1)$$

In the special case $\beta = \infty$ (which occurs whenever one of the r_i equals α , in particular for $R(t) = t^4 - K$ with $\alpha = 0$), the substitution simplifies to $z = t - \alpha$ and (1) becomes $R(t) = c(z^4 - K)$ without any $(1 - z)$ factor.

Proof. Both sides of (1) are polynomials of degree 4 in t vanishing on the four roots of R . Indeed, the right side $c(z^4 - K)$ vanishes when $z^4 = K$, i.e., when z takes the four values $\zeta K^{1/4}$ for $\zeta \in \{1, i, -1, -i\}$; these correspond, under the inverse substitution $t = (\alpha - \beta z)/(1 - z)$, to the four points r_i (since the four z -values are the $\mathbb{Z}/4$ -orbit of $K^{1/4}$, matching the cyclic action of S on the r_i). The factor $(1 - z)^4$ on the left compensates for the four poles introduced by the rational substitution. The constants c and K are then determined by matching leading coefficients and constant terms. $\square$

The transformation of the differential

A direct computation from (1) gives, for any $p \in (0, 1)$,

$$R(t)^p = \frac{c^p (z^4 - K)^p}{(1 - z)^{4p}}, \quad dt = \frac{(\alpha - \beta) dz}{(1 - z)^2}, \quad (2)$$

and hence

$$\frac{F(t) dt}{R(t)^p} = \frac{(\alpha - \beta) (1 - z)^{4p-2} F(t(z)) dz}{c^p (z^4 - K)^p}. \quad (3)$$

For $p \in \{1/4, 1/2, 3/4\}$, the exponent $4p - 2$ takes the integer values $\{-1, 0, 1\}$ respectively. We define

$$H_p(z) := \frac{(\alpha - \beta) (1 - z)^{4p-2} F(t(z))}{c^p}, \quad (4)$$

so that the integrand in z coordinates is $H_p(z) dz / (z^4 - K)^p$. (When $\beta = \infty$, the prefactor $\alpha - \beta$ is replaced by 1 and the $(1 - z)^{4p-2}$ factor is absent, exactly as for the cube-root case in [1].)

Eigenspace decomposition

The transformation $z \mapsto iz$ is a linear automorphism of order 4 on the space of rational functions of z . Its eigenspaces V_k for $k \in \mathbb{Z}/4$ are

$$V_k = \{H \in \mathbb{C}(z) : H(iz) = i^k H(z)\}.$$

A rational function $H \in V_k$ has the form $H(z) = z^k \varphi_k(z^4)$ for a unique rational φ_k . The projection $P_k : \mathbb{C}(z) \rightarrow V_k$ is given by

$$P_k H(z) = \frac{1}{4} \sum_{j=0}^3 i^{-jk} H(i^j z), \quad (5)$$

and $H = \sum_{k=0}^3 P_k H$ is the unique decomposition of H into eigencomponents.

3 The fourth-root theorem at exponent $1/4$

Reduction to algebraic curves

For the integrand $H_p(z) dz/(z^4 - K)^p$, decompose $H_p = \sum_k H_{p,k}$ with $H_{p,k} = P_k H_p \in V_k$, and write $H_{p,k}(z) = z^k \varphi_{p,k}(z^4)$. Substituting $x = z^4$, so that $dx = 4z^3 dz$, the k -th piece becomes

$$\int \frac{z^k \varphi_{p,k}(z^4) dz}{(z^4 - K)^p} = \int \frac{\varphi_{p,k}(x) \cdot z^{k-3} dx}{4(x - K)^p} = \int \frac{\varphi_{p,k}(x) dx}{4x^{(3-k)/4}(x - K)^p},$$

which we recognize as an Abelian integral on the algebraic curve

$$Y_k^{(p)}: y^4 = x^{3-k}(x - K)^{4p}. \quad (6)$$

Specifically, with $y = x^{(3-k)/4}(x - K)^p$ regarded as the relevant fourth root, the differential is

$$\omega_{p,k} = \frac{\varphi_{p,k}(x) dx}{4y}.$$

For $p = 1/4$, $4p = 1$ and the curves are

$$Y_k^{(1/4)}: y^4 = x^{3-k}(x - K), \quad k \in \{0, 1, 2, 3\}.$$

Genera

Lemma 3.1 (Genera of the obstruction curves). *For $k \in \{0, 1, 2, 3\}$ the genus of $Y_k^{(1/4)}$ is*

$$g(Y_k^{(1/4)}) = \begin{cases} 0 & \text{if } k \in \{0, 3\}, \\ 1 & \text{if } k \in \{1, 2\}. \end{cases}$$

Proof. Apply the Riemann–Hurwitz formula to the projection $Y_k^{(1/4)} \rightarrow \mathbb{P}^1$, $(x, y) \mapsto x$, of degree 4. Branch points are at $x = 0$, $x = K$, and $x = \infty$. The ramification at each branch point is

$$e_0 = \frac{4}{\gcd(4, 3-k)}, \quad e_K = \frac{4}{\gcd(4, 1)} = 4, \quad e_\infty = \frac{4}{\gcd(4, 4-k)},$$

with $\gcd(4, 3-k)$ and $\gcd(4, 4-k)$ preimages respectively. The total ramification contribution is

$$R = [4 - \gcd(4, 3-k)] + [4 - 1] + [4 - \gcd(4, 4-k)],$$

giving $2g - 2 = -8 + R$. Computing for each k :

k	$\gcd(4, 3-k)$	$\gcd(4, 4-k)$	R	$2g - 2$	g
0	1	4	$3 + 3 + 0 = 6$	-2	0
1	2	1	$2 + 3 + 3 = 8$	0	1
2	1	2	$3 + 3 + 2 = 8$	0	1
3	4	1	$0 + 3 + 3 = 6$	-2	0. $\square$

The two genus-zero parametrizations

Lemma 3.2 (Parametrization of $Y_0^{(1/4)}$). *The curve $Y_0^{(1/4)}: y^4 = x^3(x - K)$ admits the rational parametrization*

$$x = \frac{K}{1 - w^4}, \quad y = \frac{Kw}{1 - w^4}.$$

Proof. Direct verification: $y^4 = K^4 w^4 / (1 - w^4)^4$, while $x^3(x - K) = K^3 / (1 - w^4)^3 \cdot Kw^4 / (1 - w^4) = K^4 w^4 / (1 - w^4)^4$. $\square$

Lemma 3.3 (Parametrization of $Y_3^{(1/4)}$). *The curve $Y_3^{(1/4)}: y^4 = (x - K)$ admits the rational parametrization*

$$x = u^4 + K, \quad y = u.$$

Proof. Immediate. $\square$

The two lemniscate descents

Lemma 3.4 (Lemniscate descent for $Y_1^{(1/4)}$). *The curve $Y_1^{(1/4)}: y^4 = x^2(x - K)$ is birational to the elliptic curve*

$$E: y^2 = u^3 + Ku$$

via the substitution $u = y^2/x$, $x = u^2 + K$, $y^2 = u(u^2 + K)$.

Proof. With $u = y^2/x$, we have $u^2 = y^4/x^2 = x(x - K) \cdot x/x^2 = (x - K)$ from $y^4 = x^2(x - K)$, so $u^2 = x - K$ and $x = u^2 + K$. Then $y^2 = u \cdot x = u(u^2 + K) = u^3 + Ku$. The substitution $(x, y) \mapsto (u, y) = (y^2/x, y)$ is birational with inverse $u \mapsto x = u^2 + K$, y free. $\square$

Lemma 3.5 (Order-4 automorphism on $Y_2^{(1/4)}$). *The curve $Y_2^{(1/4)}: y^4 = x(x - K)$ is an elliptic curve of j -invariant 1728, hence isomorphic over $\mathbb{C}$ to the lemniscate E .*

Proof. The map $(x, y) \mapsto (x, iy)$ is an automorphism of $Y_2^{(1/4)}$ of order 4 fixing the point at infinity. Any elliptic curve admitting an automorphism of order 4 fixing the identity has j -invariant 1728 and hence has complex multiplication by $\mathbb{Z}[i]$; over $\mathbb{C}$ it is isomorphic to the unique such curve, which we may take to be $E: y^2 = u^3 + Ku$. $\square$

The main theorem at $p = 1/4$

Theorem 3.6 (Fourth-root analog of Goursat at exponent $1/4$). *Let $R(t) \in \mathbb{C}[t]$ be a quartic polynomial whose four roots are distinct and in harmonic position, S the corresponding order-4 cyclic Möbius transformation, α, β its fixed points, c, K the constants of Lemma 2.3, and $H_{1/4}$ the function defined in (4). Decompose $H_{1/4} = H_{1/4,0} + H_{1/4,1} + H_{1/4,2} + H_{1/4,3}$ via (5).*

(i) *The integral $\int F(t) dt/R(t)^{1/4}$ is elementary if and only if both $H_{1/4,1} \equiv 0$ and $H_{1/4,2} \equiv 0$.*

(ii) *When this criterion holds, the antiderivative is the sum of two pieces:*

- $J_0 = \int \varphi_{1/4,0}(K/(1 - w^4)) w^2 dw/(1 - w^4)$, a rational integral in w via Lemma 3.2 with $w = y/x = (R(t))^{1/4}/x \cdot K/y$ in canonical coordinates (i.e., w defined so that $x = K/(1 - w^4)$);
- $J_3 = \int \varphi_{1/4,3}(u^4 + K) u^2 du$, a rational integral in u via Lemma 3.3 with $u = (x - K)^{1/4}$.

(iii) *When the criterion fails, the obstruction is the non-vanishing of one or both of two non-trivial Abelian integrals on $E: y^2 = u^3 + Ku$, the lemniscate elliptic curve. Specifically, $H_{1/4,1} \not\equiv 0$ produces a non-zero differential on $Y_1^{(1/4)} \cong E$ (via Lemma 3.4), and $H_{1/4,2} \not\equiv 0$ produces a non-zero differential on $Y_2^{(1/4)} \cong E$ (via Lemma 3.5).*

Proof. The decomposition $H_{1/4} = \sum_k H_{1/4,k}$ converts the integral into the sum

$$\int \frac{F(t) dt}{R(t)^{1/4}} = \sum_{k=0}^3 \int \frac{H_{1/4,k}(z) dz}{(z^4 - K)^{1/4}} = \sum_{k=0}^3 J_k,$$

where, after the substitution $x = z^4$, each J_k is an Abelian integral on $Y_k^{(1/4)}$. By Lemma 3.1, J_0 and J_3 live on genus-0 curves, parametrized rationally by Lemmas 3.2 and 3.3, hence elementary. By Lemmas 3.4 and 3.5, J_1 and J_2 live on the elliptic curve $E: y^2 = u^3 + Ku$.

By the algebraic Liouville theorem (Theorem 2.5 of [1]), a non-zero Abelian integral on a curve of positive genus is never elementary as a function on that curve. Hence the integral $\int F dt/R^{1/4}$ is elementary if and only if both J_1 and J_2 are zero modulo elementary functions; since they live on disjoint eigenspaces and are not generically related to one another, this requires

$H_{1/4,1} = H_{1/4,2} = 0$ separately. The forward implication ($H_{1/4,1} = H_{1/4,2} = 0 \Rightarrow$ elementary) is then immediate from the parametrizations.

The reductions in (ii) follow by direct substitution. For J_0 : with $x = K/(1 - w^4)$, $y = Kw/(1 - w^4)$, we have $dx = 4Kw^3 dw/(1 - w^4)^2$, so

$$J_0 = \int \frac{\varphi_{1/4,0}(x) dx}{4y} = \int \frac{\varphi_{1/4,0}(K/(1 - w^4)) \cdot 4Kw^3 dw/(1 - w^4)^2}{4 \cdot Kw/(1 - w^4)} = \int \frac{\varphi_{1/4,0}(K/(1 - w^4)) w^2 dw}{1 - w^4}.$$

For J_3 : with $x = u^4 + K$, $y = u$, $dx = 4u^3 du$,

$$J_3 = \int \frac{\varphi_{1/4,3}(x) dx}{4y} = \int \frac{\varphi_{1/4,3}(u^4 + K) \cdot 4u^3 du}{4u} = \int \varphi_{1/4,3}(u^4 + K) u^2 du.$$

Both are rational integrals. $\square$

4 The dual theorem at exponent 3/4

The exponent $p = 3/4$ produces a dual statement, with the obstruction curves shifted by one eigenspace.

Reduction curves at $p = 3/4$

For $p = 3/4$, the curves (6) become

$$Y_k^{(3/4)}: y^4 = x^{3-k} (x - K)^3, \quad k \in \{0, 1, 2, 3\}.$$

Lemma 4.1. *The genus of $Y_k^{(3/4)}$ is*

$$g(Y_k^{(3/4)}) = \begin{cases} 1 & \text{if } k \in \{0, 1\}, \\ 0 & \text{if } k \in \{2, 3\}. \end{cases}$$

Proof. Same Riemann–Hurwitz computation as Lemma 3.1, with $b = 3$ instead of 1. The branch contribution at $x = K$ becomes $4 - \gcd(4, 3) = 3$, and the contribution at infinity is $4 - \gcd(4, (3 - k) + 3) = 4 - \gcd(4, 6 - k)$. Tabulating:

k	$\gcd(4, 3 - k)$	$\gcd(4, 6 - k)$	R	$2g - 2$	g
0	1	2	$3 + 3 + 2 = 8$	0	1
1	2	1	$2 + 3 + 3 = 8$	0	1
2	1	4	$3 + 3 + 0 = 6$	-2	0
3	4	1	$0 + 3 + 3 = 6$	-2	0. $\square$

The two new genus-zero parametrizations

Lemma 4.2 (Parametrization of $Y_2^{(3/4)}$). *The curve $Y_2^{(3/4)}: y^4 = x(x - K)^3$ admits the rational parametrization*

$$x = \frac{K s^4}{s^4 - 1}, \quad y = \frac{K s}{s^4 - 1}.$$

Proof. Direct verification: $y^4 = K^4 s^4 / (s^4 - 1)^4$, while $x(x - K)^3 = K s^4 / (s^4 - 1) \cdot [-K / (s^4 - 1)]^3 \cdot (-1)^3 = K^4 s^4 / (s^4 - 1)^4$ after simplification. $\square$

Lemma 4.3 (Parametrization of $Y_3^{(3/4)}$). *The curve $Y_3^{(3/4)}: y^4 = (x - K)^3$ admits the rational parametrization*

$$x = v^4 + K, \quad y = v^3,$$

i.e., $v = (x - K)^{1/4}$.

Proof. $y^4 = v^{12} = (v^4)^3 = (x - K)^3$. $\square$

Lemniscate descent at $p = 3/4$

The two genus-1 curves at $p = 3/4$ are also lemniscatic. The fastest way to see this is via the duality between obstructions at p and $1 - p$.

Lemma 4.4 (Bilinear duality of obstruction curves). *For each $k \in \{0, 1, 2, 3\}$, define the dual exponent and dual eigenspace by $p^* = 1 - p$ and $k^* = 2 - k \pmod{4}$. Then $Y_k^{(p)}$ and $Y_{k^*}^{(p^*)}$ are birational over $\mathbb{C}$ via the relation*

$$y y' = c x (x - K),$$

where y is the fourth root on $Y_k^{(p)}$, y' is the fourth root on $Y_{k^*}^{(p^*)}$, and c is a nonzero constant.

Proof. We have $y^4 = x^{3-k}(x-K)^{4p}$ and $(y')^4 = x^{3-k^*}(x-K)^{4p^*}$. Multiplying:

$$(y y')^4 = x^{(3-k)+(3-k^*)} \cdot (x-K)^{4(p+p^*)} = x^{6-(k+k^*)} \cdot (x-K)^4.$$

With $k+k^* = 2 \pmod{4}$, this becomes $(y y')^4 = x^4(x-K)^4 = [x(x-K)]^4$. Taking fourth roots, $y y' = c x(x-K)$ for an appropriate fourth root of unity c . $\square$

Corollary 4.5. *The curves $Y_0^{(3/4)}$ and $Y_1^{(3/4)}$ are isomorphic over $\mathbb{C}$ to $Y_2^{(1/4)}$ and $Y_1^{(1/4)}$ respectively, hence to the lemniscate $E: y^2 = u^3 + Ku$.*

Proof. Apply Lemma 4.4 with $(p, k) = (3/4, 0)$, giving $(p^*, k^*) = (1/4, 2)$, hence $Y_0^{(3/4)} \cong Y_2^{(1/4)}$. Similarly $(3/4, 1) \leftrightarrow (1/4, 1)$, hence $Y_1^{(3/4)} \cong Y_1^{(1/4)}$. By Lemmas 3.4 and 3.5, both $Y_1^{(1/4)}$ and $Y_2^{(1/4)}$ are isomorphic to E . $\square$

The main theorem at $p = 3/4$

Theorem 4.6 (Fourth-root analog of Goursat at exponent $3/4$). *Let $R(t) \in \mathbb{C}[t]$ be a quartic polynomial in harmonic configuration as in Theorem 3.6, $H_{3/4}$ the function defined in (4), decomposed as $H_{3/4} = \sum_k H_{3/4,k}$.*

- (i) *The integral $\int F(t) dt/R(t)^{3/4}$ is elementary if and only if both $H_{3/4,0} \equiv 0$ and $H_{3/4,1} \equiv 0$.*
- (ii) *When this criterion holds, the antiderivative is*

$$\begin{aligned} J_2 &= - \int \frac{\varphi_{3/4,2}(Ks^4/(s^4-1)) s^2 ds}{s^4-1} \quad (\text{Lemma 4.2}), \\ J_3 &= \int \varphi_{3/4,3}(v^4+K) dv \quad (\text{Lemma 4.3}), \end{aligned}$$

both rational in their respective parameter variables.

- (iii) *When the criterion fails, the obstructions $H_{3/4,0}$ and $H_{3/4,1}$ produce non-zero Abelian integrals on the lemniscate E via Corollary 4.5.*

Proof. The argument is identical to that of Theorem 3.6, with the eigenspace assignments shifted as in Lemma 4.1. The reductions for J_2 and J_3 follow by direct computation: for J_2 , with $x = Ks^4/(s^4-1)$, $y = Ks/(s^4-1)$, $dx = -4Ks^3 ds/(s^4-1)^2$, so

$$J_2 = \int \frac{\varphi_{3/4,2}(x) dx}{4y} = \int \frac{\varphi_{3/4,2}(Ks^4/(s^4-1)) \cdot (-4Ks^3 ds/(s^4-1)^2)}{4Ks/(s^4-1)} = - \int \frac{\varphi_{3/4,2}(Ks^4/(s^4-1)) s^2 ds}{s^4-1}.$$

For J_3 , with $x = v^4 + K$, $y = v^3$, $dx = 4v^3 dv$:

$$J_3 = \int \frac{\varphi_{3/4,3}(x) dx}{4y} = \int \frac{\varphi_{3/4,3}(v^4+K) \cdot 4v^3 dv}{4v^3} = \int \varphi_{3/4,3}(v^4+K) dv.$$

$\square$

Summary of the duality

The two main theorems exhibit the following symmetric structure:

	V_0	V_1	V_2	V_3
$p = 1/4$	elementary	<i>obstructive</i>	<i>obstructive</i>	elementary
$p = 3/4$	<i>obstructive</i>	<i>obstructive</i>	elementary	elementary

The eigenspace V_3 is always elementary (the trivial component, where the algebraic-cover structure becomes trivial); the eigenspace V_1 is always obstructive. The eigenspaces V_0 and V_2 exchange roles between $p = 1/4$ and $p = 3/4$, and Lemma 4.4 provides the explicit isomorphisms. This is the precise fourth-root analog of the duality (V_0, V_1, V_2) at $p = 1/3$ versus (V_0, V_1, V_2) at $p = 2/3$ from Corollary 4.8 of [1].

5 Worked examples on $R(t) = t^4 - 1$

The simplest harmonic quartic is $R(t) = t^4 - 1$, whose four roots $\{1, i, -1, -i\}$ are the fourth roots of unity in harmonic configuration. The order-4 Möbius transformation cycling them is $S(t) = it$, with fixed points $\alpha = 0$ and $\beta = \infty$. The substitution simplifies to $z = t$, and the canonical-form constants are $c = 1$, $K = 1$. The H -function (4) reduces to $H_p(z) = F(z)$ for both $p \in \{1/4, 3/4\}$.

Example 5.1 ($F = 1$ at $p = 1/4$: elementary). For $F(t) = 1$, $H_{1/4}(z) = 1 \in V_0$. The eigencomponents are $H_{1/4,0} = 1$ and $H_{1/4,k} = 0$ for $k \geq 1$. By Theorem 3.6(i), the integral is elementary; by (ii), the reduction is via Lemma 3.2 with $\varphi_{1/4,0}(x) = 1$:

$$\int \frac{dt}{(t^4 - 1)^{1/4}} = J_0 = \int \frac{w^2 dw}{1 - w^4}$$

where $w = (t^4 - 1)^{1/4}/t$ (one checks $x = K/(1 - w^4) = 1/(1 - w^4) = t^4$). The rational integral evaluates by partial fractions:

$$\int \frac{w^2 dw}{1 - w^4} = \frac{1}{4} \log\left(\frac{1+w}{w-1}\right) - \frac{1}{2} \arctan(w) + C.$$

Therefore

$$\int \frac{dt}{(t^4 - 1)^{1/4}} = \frac{1}{4} \log\left(\frac{t + (t^4 - 1)^{1/4}}{(t^4 - 1)^{1/4} - t}\right) - \frac{1}{2} \arctan\left(\frac{(t^4 - 1)^{1/4}}{t}\right) + C.$$

This is verifiable by direct differentiation. □

Example 5.2 ($F = t^3$ at $p = 1/4$: trivially elementary). For $F(t) = t^3$, $H_{1/4}(z) = z^3 \in V_3$, so $H_{1/4,3} = z^3$ and $\varphi_{1/4,3}(x) = 1$. By Theorem 3.6(ii), the reduction is via Lemma 3.3 with $\varphi_{1/4,3} = 1$:

$$J_3 = \int u^2 du = \frac{u^3}{3}.$$

With $u = (t^4 - 1)^{1/4}$, this gives

$$\int \frac{t^3 dt}{(t^4 - 1)^{1/4}} = \frac{(t^4 - 1)^{3/4}}{3} + C,$$

the elementary closed form one would also obtain from the trivial substitution $u = (t^4 - 1)^{1/4}$. □

Example 5.3 ($F = t$ and $F = t^2$ at $p = 1/4$: non-elementary). For $F(t) = t$, $H_{1/4}(z) = z \in V_1$, so $H_{1/4,1} = z$, $\varphi_{1/4,1}(x) = 1$. The criterion of Theorem 3.6(i) fails ($H_{1/4,1} \neq 0$), and the obstruction is the non-zero Abelian integral

$$J_1 = \int \frac{dx}{4y}, \quad y^4 = x^2(x-1),$$

i.e., a non-zero differential on $Y_1^{(1/4)}$, which is the lemniscate elliptic curve $E: y^2 = u^3 + u$ via $u = y^2/x$. Hence $\int t \, dt/(t^4 - 1)^{1/4}$ is not elementary.

Similarly, $F(t) = t^2$ gives $H_{1/4}(z) = z^2 \in V_2$, with the obstruction landing on $Y_2^{(1/4)}$: $y^4 = x(x-1)$, also lemniscate, hence non-elementary. $\square$

Example 5.4 ($F = 1$ at $p = 3/4$: non-elementary). For $F(t) = 1$, $H_{3/4}(z) = 1 \in V_0$, so $H_{3/4,0} = 1 \neq 0$. By Theorem 4.6(i), the integral $\int dt/(t^4 - 1)^{3/4}$ is non-elementary. The obstruction lives on $Y_0^{(3/4)} \cong E$ via Corollary 4.5.

This is the duality of V_0 in action: the same $F = 1$ and $R = t^4 - 1$ that gave an elementary integral at $p = 1/4$ now gives a non-elementary integral at $p = 3/4$. Geometrically, the differential $dt/(t^4 - 1)^{3/4}$ on $Y: y^4 = t^4 - 1$ has poles only at the four roots of R and is in fact a holomorphic differential of the first kind on a suitable model — hence non-elementary by the algebraic Liouville theorem. $\square$

Example 5.5 ($F = t^2$ at $p = 3/4$: elementary). For $F(t) = t^2$, $H_{3/4}(z) = z^2 \in V_2$. The eigenvectors are $H_{3/4,2} = z^2$ and $\varphi_{3/4,2}(x) = 1$. By Theorem 4.6(i)–(ii), the integral is elementary, with reduction via Lemma 4.2:

$$J_2 = -\int \frac{s^2 \, ds}{s^4 - 1} = \frac{1}{4} \log\left(\frac{1+s}{s-1}\right) - \frac{1}{2} \arctan(s) + C.$$

The parameter s is determined by $x = Ks^4/(s^4 - 1) = s^4/(s^4 - 1)$. With $x = z^4 = t^4$, this gives $s = t/(t^4 - 1)^{1/4}$. Substituting back:

$$\int \frac{t^2 \, dt}{(t^4 - 1)^{3/4}} = \frac{1}{4} \log\left(\frac{(t^4 - 1)^{1/4} + t}{t - (t^4 - 1)^{1/4}}\right) - \frac{1}{2} \arctan\left(\frac{t}{(t^4 - 1)^{1/4}}\right) + C.$$

This is the duality of Example 5.1: the same eigenspace structure (V_0 at $p = 1/4$, V_2 at $p = 3/4$) and the same closed-form template (logs and arctangent of a fourth-root substitution), with the duality $V_0 \leftrightarrow V_2$ between exponents $1/4$ and $3/4$ now displayed concretely. $\square$

Example 5.6 ($F = t^3$ at $p = 3/4$: trivially elementary). For $F(t) = t^3$, $H_{3/4}(z) = z^3 \in V_3$, so $H_{3/4,3} = z^3$, $\varphi_{3/4,3}(x) = 1$. By Theorem 4.6(ii):

$$J_3 = \int dv = v.$$

With $v = (t^4 - 1)^{1/4}$ this gives

$$\int \frac{t^3 \, dt}{(t^4 - 1)^{3/4}} = (t^4 - 1)^{1/4} + C,$$

the elementary closed form one would obtain from the substitution $v = (t^4 - 1)^{1/4}$ directly. $\square$

Example 5.7 ($F = t$ at $p = 3/4$: non-elementary). For $F(t) = t$, $H_{3/4}(z) = z \in V_1$. By Theorem 4.6(i), the integral is non-elementary, with obstruction on $Y_1^{(3/4)} \cong E$. This continues to demonstrate that V_1 is always obstructive, regardless of p . $\square$

6 Concluding remarks

Lemniscate rigidity.

Across both Theorems 3.6 and 4.6, every genus-1 obstruction curve is isomorphic over $\mathbb{C}$ to the same elliptic curve $E: y^2 = u^3 + Ku$, the lemniscate. This rigidity is forced by the underlying $\mathbb{Z}/4$ -symmetry: any elliptic curve admitting an automorphism of order 4 has j -invariant 1728 and complex multiplication by $\mathbb{Z}[i]$. The four obstruction curves $Y_1^{(1/4)}, Y_2^{(1/4)}, Y_0^{(3/4)}, Y_1^{(3/4)}$ are pairwise isomorphic via the bilinear duality $yy' = cx(x - K)$ of Lemma 4.4, all collapsing onto the single lemniscate E regardless of which exponent or eigenspace produced them. This rigidity is genuinely special to $n = 4$; for $n = 5$, the three interior eigenspaces lead to three independent genus-2 hyperelliptic curves with no analogous unifying isomorphism. (See the concluding remarks of [1] for the genus picture at $n \geq 5$.)

The Möbius automorphism.

Lemma 3.5 (the order-4 automorphism on $Y_2^{(1/4)}$) gives the cleanest proof that this curve is lemniscatic, and is itself a manifestation of the $\mathbb{Z}/4$ -symmetry. A parallel computation shows that $Y_1^{(3/4)}$ admits the order-4 automorphism $(x, y) \mapsto (x, iy)$ as well, since $y^4 = x^2(x - K)^3$ is invariant under $y \mapsto iy$. The other two genus-1 curves ($Y_1^{(1/4)}$ and $Y_0^{(3/4)}$) acquire their lemniscate structure indirectly, via the substitution $u = y^2/x$ on $Y_1^{(1/4)}$ and via the duality of Lemma 4.4 on $Y_0^{(3/4)}$.

Comparison with $n = 3$.

Theorems 3.6 and 4.6 parallel Theorem 4.3 and Corollary 4.8 of [1], but with three structural differences:

1. **Two simultaneous obstructions** instead of one. At $n = 3$, exactly one eigenspace (V_1) obstructed elementarity; at $n = 4$, two eigenspaces obstruct simultaneously. The reason is the genus formula $g(Y_k) = (n + 1 - \gcd(n, k) - \gcd(n, k + 1))/2$ from [1]: for $n = 3$, only $k = 1$ has gcd values both equal to 1; for $n = 4$, both $k = 1$ and $k = 2$ do.
2. **An existence hypothesis** on R . The cube-root case applied to every cubic R with simple roots; the fourth-root case requires the harmonic condition (cross-ratio -1). Equivalently, the moduli space of harmonic quartics is one-dimensional inside the two-dimensional moduli of all quartics with simple roots.
3. **Lemniscate rigidity.** The $n = 3$ case landed on a single elliptic curve $y^3 = x(x - K)$ (the cube-root analog of the Weierstrass form), whose j -invariant depends on K . The $n = 4$ case lands on $y^2 = u^3 + Ku$ with $j = 1728$ *regardless* of K . The genus-1 moduli is therefore 0-dimensional in the $n = 4$ case (modulo K scaling), in contrast to 1-dimensional in the $n = 3$ case.

Algorithmic perspective.

The diagnostic algorithm extending `IntegrateGoursat` to the cases $p \in \{1/4, 3/4\}$ requires:

1. A test for the harmonic condition on the four roots of R (computing the cross-ratio and checking it equals -1).
2. Computation of the order-4 Möbius S , its fixed points α, β , and the canonical-form constants c, K from Lemma 2.3.

3. Computation of $H_p(z)$ from (4) and the four eigenspace projections $H_{p,k}$ via (5).
4. Application of Theorem 3.6 or 4.6 as appropriate.

The resulting test is purely algebraic, polynomial-time in the degrees of R and F , and yields a constructive certificate of (non-)elementarity. We give an implementation in Appendix A, extending the `IntegrateGoursat` package of [1] to accept the new exponents.

Future directions.

The next natural extension is to higher cyclic radicals $n \geq 5$ under the corresponding harmonic-tuple conditions, where the obstruction curves are higher-genus hyperelliptic and superelliptic curves. The genus formula from [1] predicts the qualitative shape of these obstructions, but the lemniscate-style rigidity is lost: at $n = 5$ for example, the three interior eigenspaces produce three genuinely different genus-2 curves with no $\mathbb{Z}/5$ -induced isomorphisms among them. A separate but related direction is the case of $\sqrt{R}$ for sextic R , where the relevant curves are hyperelliptic of genus 2 and the symmetry classification involves subgroups of $\text{Aut}(\mathbb{P}^1)$ of various types (single involution, V_4 , $\mathbb{Z}/3$, D_3 , etc.) acting on the six roots. We hope to return to both directions in subsequent work.

A Mathematica implementation

We extend the `IntegrateGoursat` package of [1] with two new branches, `QuarticTest14` and `QuarticTest34`, implementing the algorithms of Theorems 3.6 and 4.6 respectively. The unified entry point now accepts $p \in \{1/2, 1/3, 2/3, 1/4, 3/4\}$:

$$\text{IntegrateGoursat}[F, R, t, p], \quad p \in \{1/2, 1/3, 2/3, 1/4, 3/4\}.$$

For $p \in \{1/4, 3/4\}$, the package first tests the harmonic condition on the roots of R , returning "non-harmonic" if it fails (in which case the present theory does not apply); otherwise it computes the order-4 Möbius transformation S , its fixed points, the canonical-form constants c, K , and the four eigenspace projections of H_p from (4), returning "status" $\rightarrow$ "elementary" or "non-elementary" according to the criterion of the appropriate theorem.

Source code (extension only)

```
(* == Fourth-root case (Theorems 3.7 and 4.7) == *)

(* Test if four points are in harmonic position. *)
HarmonicConfigQ[roots_List] :=
Module[{cr, r = roots},
  If[Length[r] != 4, Return[False]];
  cr = ((r[[1]] - r[[2]])(r[[3]] - r[[4]])/
    ((r[[1]] - r[[4]])(r[[3]] - r[[2]]));
  PossibleZeroQ[Simplify[cr + 1]];

(* Order-4 Möbius transformation cycling the 4 roots. *)
QuarticMöbius[roots_List, t_] :=
Module[{r = roots, A, B, C, D, sol},
  sol = First[Solve[
    {(A r[[1]] + B)/(C r[[1]] + D) == r[[2]],
     (A r[[2]] + B)/(C r[[2]] + D) == r[[3]],
     (A r[[3]] + B)/(C r[[3]] + D) == r[[4]],
     D == 1}, {A, B, C, D}]];
  Together[(A t + B)/(C t + 1) /. sol];
```

```

(* Eigenspace projection P_k under z -> i*z. *)
EigenProjection4[H_, z_, k_Integer] :=
  Together[Simplify[
    (H + I^(-k) (H /. z -> I z) +
     I^(-2 k) (H /. z -> -z) +
     I^(-3 k) (H /. z -> -I z))/4]];

(* Convert a function divisible by z^k into a function of x = z^4. *)
ToFunctionOfFourth[H_, z_, x_] :=
  If[PossibleZeroQ[Together[H]], 0,
    Module[{num, den, dN, dD},
      {num, den} = Through[{Numerator, Denominator}[Together[H]]];
      dN = Exponent[num, z]; dD = Exponent[den, z];
      Sum[Coefficient[num, z, 4 k] x^k, {k, 0, dN/4}]/
      Sum[Coefficient[den, z, 4 k] x^k, {k, 0, dD/4}]]];

(* Common setup for both p = 1/4 and p = 3/4. *)
QuarticSetup[F_, R_, t_, p_] :=
  Module[{roots, S, fps, alpha, beta, z, tz, Rz, cval, K, H},
    roots = DeleteDuplicates[t /. Solve[R == 0, t]];
    If[Length[roots] != 4, Return[$Failed]];
    If[!HarmonicConfigQ[roots],
      Return[<|"status" -> "non-harmonic"|>]];
    S = QuarticMobius[roots, t];
    fps = t /. Solve[S == t, t];
    {alpha, beta} = If[Length[fps] == 1, {fps[[1]], Infinity}, fps];
    tz = If[beta === Infinity, alpha + z, (alpha - beta z)/(1 - z)];
    Rz = If[beta === Infinity,
      Cancel[Together[R /. t -> tz]],
      Cancel[Together[(R /. t -> tz) (1 - z)^4]]];
    cval = Coefficient[Expand[Rz], z, 4];
    K = -Coefficient[Expand[Rz], z, 0]/cval;
    H = Together[
      If[beta === Infinity,
        (F /. t -> tz)/cval^p,
        (alpha - beta) (1 - z)^(4 p - 2) (F /. t -> tz)/cval^p];
      <|"S" -> S, "alpha" -> alpha, "beta" -> beta,
      "K" -> K, "c" -> cval, "H" -> H|>];

  QuarticTest14[F_, R_, t_] :=
    Module[{setup, H, H0, H1, H2, H3, w, u, x,
      phi0, phi3, J0, J3},
      setup = QuarticSetup[F, R, t, 1/4];
      If[!AssociationQ[setup] || setup["status"] === "non-harmonic",
        Return[setup]];
      H = setup["H"];
      {H0, H1, H2, H3} =
        EigenProjection4[H, z, #] & /@ {0, 1, 2, 3};
      If[!PossibleZeroQ[Simplify[H1]] || !PossibleZeroQ[Simplify[H2]],
        Return[Append[setup,
          <|"status" -> "non-elementary",
          "obstruction1" -> H1, "obstruction2" -> H2|>]]];
      phi0 = ToFunctionOfFourth[H0, z, x];
      phi3 = ToFunctionOfFourth[Together[H3/z^3], z, x];
      J0 = Together[
        (phi0 /. x -> setup["K"]/(1 - w^4)) w^2/(1 - w^4)];
      J3 = Together[(phi3 /. x -> u^4 + setup["K"]) u^2];
      Append[setup, <|"status" -> "elementary",
        "H0" -> H0, "H3" -> H3,
        "J0integrand" -> J0, "J3integrand" -> J3|>]];

  QuarticTest34[F_, R_, t_] :=
    Module[{setup, H, H0, H1, H2, H3, s, v, x,

```

```

    phi2, phi3, J2, J3},
    setup = QuarticSetup[F, R, t, 3/4];
    If[!AssociationQ[setup] || setup["status"] === "non-harmonic",
      Return[setup]];
    H = setup["H"];
    {H0, H1, H2, H3} =
      EigenProjection4[H, z, #] & /@ {0, 1, 2, 3};
    If[!PossibleZeroQ[Simplify[H0]] || !PossibleZeroQ[Simplify[H1]],
      Return[Append[setup,
        <|"status" -> "non-elementary",
          "obstruction0" -> H0, "obstruction1" -> H1|>]]];
    phi2 = ToFunctionOfFourth[Together[H2/z^2], z, x];
    phi3 = ToFunctionOfFourth[Together[H3/z^3], z, x];
    J2 = Together[
      -(phi2 /. x -> setup["K"] s^4/(s^4 - 1)) s^2/(s^4 - 1)];
    J3 = Together[phi3 /. x -> v^4 + setup["K"]];
    Append[setup, <|"status" -> "elementary",
      "H2" -> H2, "H3" -> H3,
      "J2integrand" -> J2, "J3integrand" -> J3|>]];

(* === Updated entry point === *)

IntegrateGoursat[F_, R_, t_, p_:1/2] :=
  Switch[p,
    1/2, GoursatTest[F, R, t],
    1/3, CubicTest[F, R, t],
    2/3, CubicTest23[F, R, t],
    1/4, QuarticTest14[F, R, t],
    3/4, QuarticTest34[F, R, t],
    _, "Only p = 1/2, 1/3, 2/3, 1/4, 3/4 are supported"];

```

Sample sessions

```
In[]:= IntegrateGoursat[1, t^4 - 1, t, 1/4]
```

returns "status" $\rightarrow$ "elementary", $K = 1$, $H_0 = 1$, $H_3 = 0$, with $J_0\text{integrand} = w^2/(1 - w^4)$ and $J_3\text{integrand} = 0$. By partial fractions, the antiderivative is the closed form of Example 5.1.

```
In[]:= IntegrateGoursat[t, t^4 - 1, t, 1/4]
In[]:= IntegrateGoursat[t^2, t^4 - 1, t, 1/4]
```

both return "status" $\rightarrow$ "non-elementary": the first with obstruction $H_1 = z$, the second with $H_2 = z^2$. Both witness Abelian integrals on the lemniscate $y^2 = u^3 + u$.

```
In[]:= IntegrateGoursat[1, t^4 - 1, t, 3/4]
```

returns "status" $\rightarrow$ "non-elementary" with obstruction $H_0 = 1$. The same input that was elementary at exponent 1/4 is non-elementary at 3/4, as predicted by the $V_0 \leftrightarrow V_2$ duality.

```
In[]:= IntegrateGoursat[t^2, t^4 - 1, t, 3/4]
```

returns "status" $\rightarrow$ "elementary", $K = 1$, $H_2 = z^2$, $H_3 = 0$, with $J_2\text{integrand} = -s^2/(s^4 - 1)$ — the same partial-fraction template as Example 5.1, dualised by the eigenspace shift.

The harmonic test `HarmonicConfigQ` prevents misapplication: calling `IntegrateGoursat[F, R, t, 1/4]` on a non-harmonic quartic such as $R = (t-1)(t-2)(t-3)(t-5)$ returns "non-harmonic", signalling that the present theory does not apply (although the integral may still be elementary by other methods, just as a non-harmonic Goursat integrand may be elementary by Theorem 3.5 of [1] or by the Risch–Trager–Bronstein algorithm).

References

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